

WASP-52b

R-0486

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230924 · created 2026-07-20T08:33:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460211.8774 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1872 +/- 0.0073
Transit depth (Rp/Rs)^2	3.5 +/- 0.27 [%]
Orbital Inclination (inc)	87.52 +/- 0.97
Airmass coefficient 1 (a1)	0.995 +/- 0.011
Airmass coefficient 2 (a2)	0.0043 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#1 - [265, 241]
Optimal Aperture	4.72
Optimal Annulus	15.61
Transit Duration (day)	0.0863 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1872 $\pm$ 0.0073 vs 0.1646 (deviation 3.1 σ)
Duration fit vs published	0.0863 d vs 0.0758 d (deviation 4.03 σ)
Mid-time O-C	-2.2 min
Depth SNR	25.6
Residual scatter	1.11 %

Light curve

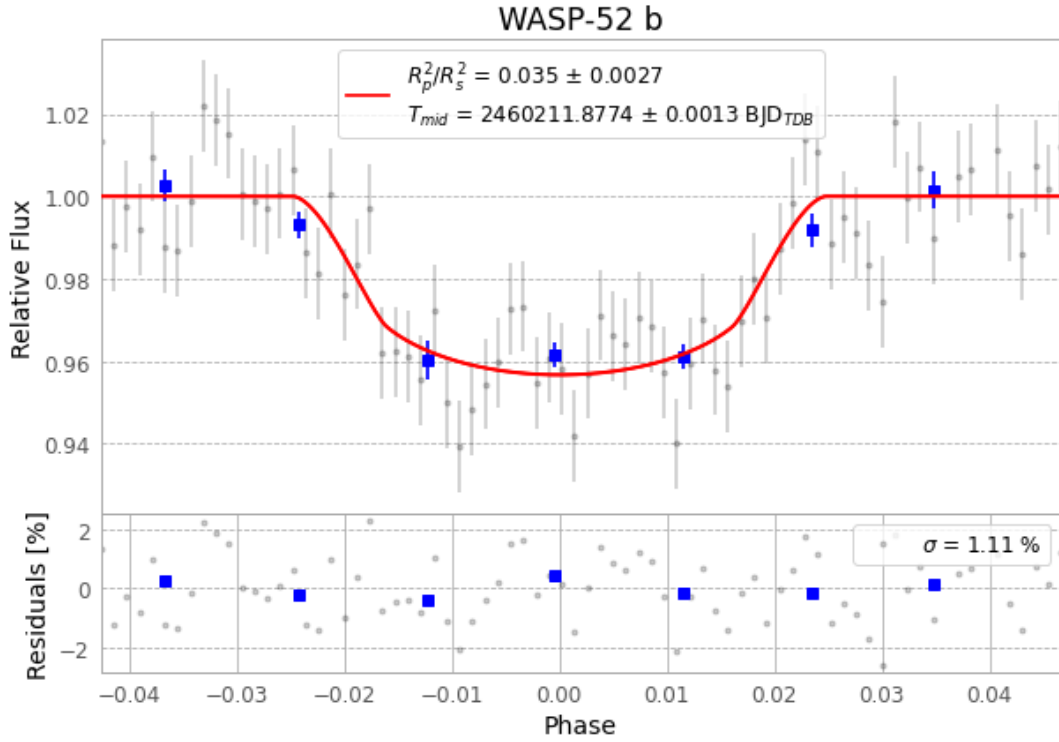


Figure 1 — FinalLightCurve_WASP-52 b_2023-09-24.png (SHA-256 064f90d7b5c9c78a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 170], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4b31c87a7f950e5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

76 FITS frames (50 MB), WASP-52230924070615.FITS → WASP-52230924105115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230924.log (a20dc4814cce...), FinalLightCurve_WASP-52 b_2023-09-24.png (064f90d7b5c9...), FinalLightCurve_WASP-52 b_2023-09-24.csv (b5172c116c21...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 08:33Z.